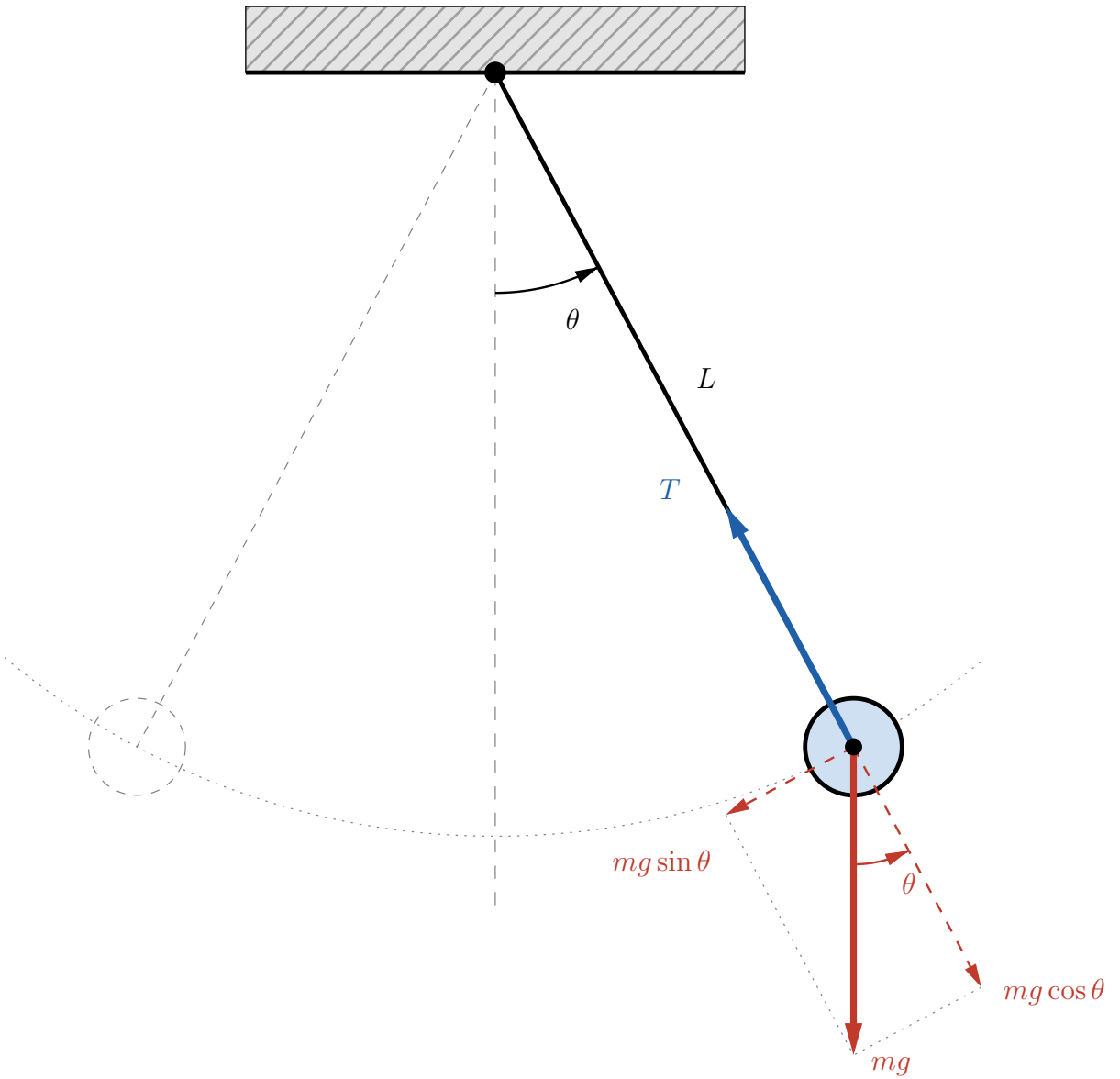
Simple pendulum

Forces on the bob at the turning point of its swing



At the turning point ($v = 0$):

$T = mg\cos\theta$

$mL\,\dfrac{d^2\theta}{dt^2} = -mg\sin\theta$

Small angles: $\omega = \sqrt{g/L}$